

KISUMU, Kenya

WMO: 637080

Lat: 0.086S

Lon: 34.729E

Elev: 3965

StdP: 12.71

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	60.4	61.6	45.8	52.7	81.6	49.2	60.0	78.5	22.3	75.9	19.2	77.0	2.1	60	0.298

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	23.0	91.7	65.9	89.8	66.8	88.0	67.4	72.8	82.4	72.0	81.9	71.4	81.4	13.0	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.0	128.0	77.5	69.1	124.1	76.5	68.3	120.7	75.6	39.4	82.5	38.6	82.6	38.0	81.5	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.0	19.2	17.0	DB	53.8	95.7	3.7	2.1	51.1	97.3	48.9	98.5	46.8	99.7	44.1	101.2	(4)
(5)			WB	51.1	77.1	2.2	3.1	49.6	79.4	48.3	81.2	47.0	82.9	45.4	85.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	74.6	75.3	76.5	76.3	74.8	73.8	73.1	72.8	73.5	74.4	75.2	74.5	74.7	(6)	
	DBStd	2.29	2.26	2.44	2.65	1.88	1.45	1.54	1.87	1.79	1.95	1.98	1.89	2.13	(7)	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD50	8967	784	741	816	743	739	693	708	728	731	782	736	766	(10)	
	CDD65	3492	319	321	351	293	274	243	243	263	281	317	286	301	(11)	
	CDH74	24416	2656	2936	2865	1830	1525	1413	1461	1629	1899	2149	1816	2237	(12)	
	CDH80	7292	956	1225	1092	459	275	237	280	383	570	661	466	688	(13)	
(14) Wind	WSAvg	5.9	6.8	7.2	7.0	5.6	4.9	4.9	5.2	5.6	6.0	5.7	5.4	6.1	(14)	
(15) Precipitation	PrecAvg	52.5	2.4	3.8	5.6	8.3	6.8	2.9	2.4	4.3	3.6	3.1	5.4	4.1	(15)	
	PrecMax	64.4	9.6	10.6	11.3	14.1	12.6	5.5	5.5	8.9	6.6	7.1	18.3	10.9	(16)	
	PrecMin	37.4	0.2	0.4	0.0	2.4	3.0	1.0	0.5	1.1	0.3	0.7	1.6	0.9	(17)	
	PrecStd	7.9	2.2	2.8	3.1	3.1	2.6	1.1	1.4	2.3	1.9	1.8	3.9	2.3	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.1	94.9	94.8	89.4	86.0	85.5	86.2	87.8	90.8	89.9	88.7	91.7	(19)	
		MCWB	64.6	64.4	65.0	68.4	69.2	67.2	66.3	65.8	66.6	67.1	67.2	65.6	(20)	
	2%	DB	89.8	91.8	91.1	86.3	84.3	84.1	84.4	85.8	87.6	87.9	86.2	88.1	(21)	
		MCWB	66.0	65.4	67.3	69.6	69.3	68.0	66.4	66.9	67.0	67.6	68.1	66.5	(22)	
	5%	DB	87.7	89.6	88.2	84.4	82.8	82.6	82.8	84.1	85.6	86.0	84.5	86.0	(23)	
		MCWB	66.8	66.7	68.5	69.8	69.5	68.1	67.0	67.2	67.6	68.1	68.5	67.3	(24)	
	10%	DB	85.6	87.4	86.0	82.6	81.5	81.0	81.6	82.3	83.5	84.1	82.7	84.1	(25)	
		MCWB	67.8	67.5	68.9	69.8	69.4	68.2	67.2	67.6	68.1	68.2	68.8	67.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.8	73.0	73.4	73.6	72.6	71.7	70.7	71.3	72.2	72.4	73.3	72.6	(27)	
		MCDB	82.6	85.1	84.0	82.7	81.0	81.2	80.9	81.4	82.6	81.8	81.4	81.1	(28)	
	2%	WB	71.5	71.5	72.2	72.2	71.5	70.4	69.6	69.9	70.9	71.0	71.3	71.3	(29)	
		MCDB	81.8	83.2	82.9	81.6	80.5	80.1	80.0	81.1	81.7	81.6	80.9	80.6	(30)	
	5%	WB	70.5	70.5	71.3	71.4	70.7	69.6	68.7	69.0	69.8	70.0	70.3	70.3	(31)	
		MCDB	81.0	82.1	81.9	80.8	79.7	79.4	79.1	80.1	81.1	81.0	80.1	79.7	(32)	
	10%	WB	69.6	69.7	70.4	70.5	70.0	68.9	67.9	68.1	68.7	69.1	69.5	69.5	(33)	
		MCDB	80.1	81.1	80.7	79.6	79.0	78.4	78.2	79.2	79.9	79.7	79.1	78.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	21.8	23.0	21.0	18.0	17.4	18.3	19.0	19.7	20.9	20.3	19.0	19.7	(35)	
		MCDBR	25.6	27.3	24.6	20.3	19.0	19.9	21.0	22.0	23.9	22.8	21.1	23.4	(36)	
	5% WB	MCWBR	10.9	11.2	9.9	7.9	7.6	8.0	8.0	8.3	9.1	8.3	8.1	9.7	(37)	
		MCWBR	21.0	22.0	20.8	18.2	17.3	17.8	18.7	19.7	20.5	19.8	18.8	19.0	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.385	0.409	0.410	0.388	0.388	0.447	0.461	0.438	0.405	0.392	0.379	0.382	(40)	
		taud	2.428	2.353	2.374	2.466	2.462	2.261	2.202	2.274	2.372	2.412	2.469	2.448	(41)	
	Ebn at Noon		299	293	290	290	283	261	259	272	287	294	299	299	(42)	
		Edn at Noon	38	42	41	36	35	42	45	43	40	39	36	37	(43)	
(44) All-Sky Solar Radiation	RadAvg		1972	2072	2041	1879	1828	1721	1692	1779	1951	1910	1815	1879	(44)	
	RadStd		131	100	135	114	63	89	85	79	74	78	102	115	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page